
Latent Space Cartography Applied to Wikidata: Relational Displacement Analysis Reveals a Silent Tokenizer Defect in mxbai-embed-large

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github.com/EmmaLeonhart/latent-space-cartography

Abstract

We apply latent space cartography — the systematic mapping of structure in pre-trained embedding spaces (Liu et al., 2019) — to three general-purpose text embedding models using Wikidata knowledge graph triples as probes. The method is a standard application of TransE-style relational displacement analysis (Bordes et al., 2013) to frozen (non-KGE) embeddings: given any embedding model and any knowledge base, it discovers which relations manifest as consistent vector displacements and which do not. Applied to mxbai-embed-large (1024-dim), nomic-embed-text (768-dim), and all-minilm (384-dim), the procedure identifies 30 relations that are consistent across all three models, confirming that these are properties of the semantic relationships rather than artifacts of any single model. A correlation between geometric consistency and prediction accuracy ($r = 0.861$, 95% CI [0.773, 0.926]) reproduces across models, meaning the consistency metric predicts which discovered operations will be useful without held-out evaluation.

The primary empirical finding is a previously unreported defect in mxbai-embed-large: 147,687 cross-entity embedding pairs at cosine similarity ≥ 0.95 , caused by WordPiece [UNK] token dominance. When the tokenizer encounters characters with diacritical marks (ō, ū, ī, ï, ș, ț, etc.), these characters are absent from the WordPiece vocabulary and are replaced with [UNK] tokens. For short input strings where most characters are out-of-vocabulary, the resulting embedding is governed by the [UNK] token representation rather than by the actual text content, causing all such inputs to converge to a single region of the embedding space. Empirically, "Hokkaidō" has cosine similarity 1.0 with "Éire", "Djazair", and "Filastīn" — completely unrelated words in different languages — while having cosine similarity of only 0.45 with its own ASCII equivalent "Hokkaido". These collisions occupy the densest regions of the embedding space (71% in the densest quartile). This defect is silent: it affects any RAG system, semantic search engine, or knowledge graph application using mxbai-embed-large with non-ASCII input, and standard benchmarks (MTEB, etc.) do not test for it. The method, all code, and all data are publicly available.

1 Introduction

That embedding spaces encode relational structure as vector arithmetic is well established. The word2vec analogy $\text{king} - \text{man} + \text{woman} \approx \text{queen}$ (Mikolov et al., 2013) demonstrated this for distributional word embeddings. TransE (Bordes et al., 2013) formalized the insight for knowledge graphs, training embeddings such that $h + r \approx t$ for each triple (head, relation, tail). Subsequent work introduced rotations (RotatE; Sun et al., 2019), complex-valued embeddings (ComplEx; Trouillon et al., 2016), geometric constraints for hierarchical relations (box embeddings; Vilnis et al., 2018), and extensive theoretical analysis of which relation types admit which geometric representations (e.g., Wang et al., 2014; Kazemi & Poole, 2018).

The KGE research program is *constructive*: it builds embedding spaces optimized for relational reasoning. A complementary *cartographic* approach — mapping the structure that pre-trained spaces

already encode — has been explored through visual analysis tools (Liu et al., 2019) and probing classifiers (Conneau et al., 2018; Hewitt & Manning, 2019), but these techniques are typically applied to answer specific hypotheses about specific models. Systematic relational mapping across all predicates in a knowledge base, applied to frozen general-purpose embeddings, remains underexplored.

We apply standard TransE-style relational displacement analysis to frozen text embeddings, systematically sweeping over all predicates in a Wikidata knowledge graph. The procedure is not methodologically novel — it packages known techniques (displacement consistency, leave-one-out evaluation) into a replicable pipeline. What is novel is what the pipeline found when applied to a domain that standard benchmarks do not cover.

The paper has three contributions:

1. **Cross-model relational mapping.** Applied to three models (mxbai-embed-large, nomic-embed-text, all-minilm), the procedure identifies 30 relations that manifest as consistent displacements across all three — confirming that the mapped structure is a property of the semantic relationships, not any particular model. A correlation between consistency and prediction accuracy ($r = 0.861$) means the consistency metric is self-calibrating.
2. **Discovery of a silent tokenizer defect.** The same procedure, applied to a domain-specific seed (Engishiki, a Japanese historical text), surfaced a large-scale defect in mxbai-embed-large: 147,687 cross-entity embedding pairs at cosine ≥ 0.95 . The root cause is WordPiece [UNK] token dominance — characters with diacritical marks are absent from the tokenizer vocabulary, replaced with [UNK], and when [UNK] tokens dominate a short input, the embedding converges to a single attractor region regardless of text content. This defect is silent, has existed for years in a widely-deployed model, and affects any downstream system processing non-ASCII input.
3. **Empirical evidence of the defect mechanism.** We provide controlled test pairs demonstrating that the diacritical version of a word (e.g., "Hokkaidō") is more similar to an unrelated diacritical word ("Éire", cosine 1.0) than to its own ASCII equivalent ("Hokkaido", cosine 0.45). This rules out diacritic *stripping* as the mechanism and points to [UNK] token *dominance* — the model does not see the stripped version; it sees a sequence of unknown tokens that overwhelms the actual text signal.

1.1 Key Findings

1. **Relational displacement generalizes across models.** Of 159 predicates tested (≥ 10 triples each), 86 produce consistent displacement vectors in mxbai-embed-large, with 30 universal across all three models. Functional (many-to-one) relations encode as consistent displacements; symmetric relations do not — matching the predictions of the KGE literature (Wang et al., 2014).
2. **Consistency predicts accuracy.** The correlation between geometric consistency and prediction accuracy ($r = 0.861$, 95% CI [0.773, 0.926]) means the consistency metric functions as a self-calibrating quality indicator. This correlation is not tautological: consistency is computed over all triples, while MRR uses leave-one-out evaluation where each prediction excludes the test triple.
3. **A silent defect in mxbai-embed-large.** The procedure revealed 147,687 cross-entity embedding pairs at cosine ≥ 0.95 , caused by [UNK] token dominance in the WordPiece tokenizer. The defect collapses all short diacritical strings — regardless of language, script, or meaning — into a single dense region of the space. Controlled pairs confirm the mechanism: "Hokkaidō" $\leftrightarrow$ "Éire" = 1.0 cosine, "Hokkaidō" $\leftrightarrow$ "Hokkaido" = 0.45 cosine.
4. **The defect is silent and systemic.** Standard benchmarks (MTEB, etc.) do not test diacritic-rich input at scale. The defect has existed for years in a widely-deployed model. Any RAG system or semantic search using mxbai-embed-large will silently fail on queries containing diacritical marks — returning results from the [UNK] attractor region instead of semantically relevant results.
5. **Domain-specific seeds expose domain-specific failures.** The Engishiki seed (a Japanese historical text) naturally reaches romanized non-Latin terminology that standard bench-

marks never touch. This is not a limitation but an experimental design choice: different seeds probe different regions of the embedding space.

2 Related Work

2.1 Knowledge Graph Embedding

TransE (Bordes et al., 2013) established that relations can be modeled as translations ($h + r \approx t$) in learned embedding spaces. Subsequent work analyzed which relation types each model can represent: TransE handles antisymmetric and compositional relations but cannot model symmetric ones; RotatE (Sun et al., 2019) handles symmetry via rotation; ComplEx (Trouillon et al., 2016) handles symmetry and antisymmetry via complex-valued embeddings. Wang et al. (2014) and Kazemi & Poole (2018) provided systematic analyses of the relation type expressiveness of different KGE architectures. Our work does not introduce a new embedding method but applies the known displacement test systematically to frozen general-purpose (non-KGE) embedding spaces.

2.2 Word Embedding Analogies

Mikolov et al. (2013) showed that $\text{king} - \text{man} + \text{woman} \approx \text{queen}$ holds in word2vec. Subsequent work (Linzen, 2016; Rogers et al., 2017; Schluter, 2018) showed these analogies are less robust than initially claimed, often reflecting frequency biases and dataset artifacts. Ethayarajh et al. (2019) formalized the conditions under which analogy recovery succeeds, showing it requires the relation to be approximately linear and low-rank in the embedding space. Our work is consistent with these findings: the relations we recover are exactly those that satisfy the linearity condition (functional, bijective), and those that fail are those the theory predicts will fail (symmetric, many-to-many).

2.3 Latent Space Cartography

Liu et al. (2019) introduced *latent space cartography* as a visual analysis framework for interpreting vector space embeddings, enabling discovery of relationships, definition of attribute vectors, and verification of findings across latent spaces. Their work demonstrated the cartographic approach on image generation models, cancer transcriptomes, and word embedding benchmarks. Our work extends this cartographic paradigm to systematic relational displacement analysis: rather than visual exploration, we sweep over all predicates in a knowledge graph and characterize which relations encode as consistent vector arithmetic. The individual techniques (displacement consistency, leave-one-out evaluation) are standard; we apply them systematically as a mapping procedure.

2.4 Neurosymbolic Integration

Logic Tensor Networks (Serafini & Garcez, 2016), Neural Theorem Provers (Rocktäschel & Riedel, 2017), and DeepProbLog (Manhaeve et al., 2018) integrate logical reasoning into neural architectures. These constructive approaches build systems that reason logically. Our work maps what relational structure existing spaces already encode, rather than building new systems to produce it.

2.5 Probing and Representation Analysis

Probing classifiers (Conneau et al., 2018; Hewitt & Manning, 2019) test what linguistic properties are encoded in learned representations. Our displacement consistency metric is analogous to a probe, but operates at the relational level and uses vector arithmetic rather than learned classifiers. Rather than testing specific hypotheses, we sweep over all available predicates in a knowledge base.

2.6 Embedding Defects and Failure Modes

The glitch token phenomenon (Li et al., 2024) documents poorly trained embeddings for low-frequency tokens in LLMs. Our collision finding extends this to sentence-embedding models, showing that entire *classes* of input (romanized non-Latin scripts, diacritical text) collapse into near-identical regions. Systematic relational probing detects these defects as a byproduct, providing a practical auditing tool for embedding quality.

2.7 Tokenizer-Induced Information Loss

WordPiece (Schuster & Nakajima, 2012) and BPE (Sennrich et al., 2016) tokenizers are known to struggle with out-of-vocabulary and non-Latin text. Rust et al. (2021) showed that tokenizer quality strongly predicts downstream multilingual model performance. Systematic relational probing provides a way to detect these failures geometrically: by probing a specific domain via BFS traversal, tokenizer-induced information loss becomes visible as large-scale embedding collisions.

3 Method

3.1 Problem Formulation

Given: - An embedding function $f : \text{Text} \rightarrow \mathbb{R}^d$ (any text embedding model) - A knowledge base $\mathcal{K} = \{(s, p, o)\}$ of subject-predicate-object triples

Find: The subset of predicates $P^* \subseteq P$ whose triples manifest as consistent displacement vectors in the embedding space.

Definition (Relational Displacement). For a triple $(s, p, o) \in \mathcal{K}$, the *relational displacement* is the vector $\mathbf{g}_{s,p,o} = f(o) - f(s)$, connecting the subject’s embedding to the object’s embedding. This is the standard TransE formulation applied without training.

Definition (Displacement Consistency). For a predicate p with triples $\{(s_1, p, o_1), \dots, (s_n, p, o_n)\}$, the *mean displacement* is $\mathbf{d}_p = \frac{1}{n} \sum_{i=1}^n \mathbf{g}_{s_i,p,o_i}$. The *consistency* of p is the mean cosine alignment of individual displacements with the mean:

$$\text{consistency}(p) = \frac{1}{n} \sum_{i=1}^n \cos(\mathbf{g}_{s_i,p,o_i}, \mathbf{d}_p)$$

A predicate with consistency > 0.5 encodes as a **consistent relational displacement**: its triples are approximated by a single vector operation. This threshold is not novel — it corresponds to the standard criterion for meaningful directional agreement in high-dimensional spaces.

3.2 Data Pipeline: Knowledge Graph Traversal as Probing Strategy

The key methodological choice is using **breadth-first search through an existing knowledge graph** to generate embedding probes. This inverts the typical KGE pipeline. Standard KGE methods start with an embedding space and train it to encode known relations. Our method starts with a knowledge graph and uses its structure to *probe* an existing embedding space — the graph tells us which pairs of entities *should* be related, and the embedding tells us whether that relationship manifests geometrically.

BFS from a seed entity is not merely a data collection convenience. It is a **directed probing strategy**: by choosing a seed in a specific domain (e.g., Englishiki, a Japanese historical text), the traversal naturally reaches the entities and terminology that are most relevant to that domain. This means the method systematically tests the embedding space in regions where it may be weakest — regions populated by obscure, non-Latin, or domain-specific terminology that standard benchmarks never touch. A seed in Japanese history pulls in romanized shrine names, historical figures with diacritical marks, and linked entities from Arabic, Irish, and indigenous-language Wikipedia articles. A seed in geography or biography would probe different regions. The choice of seed controls *where* the map is drawn.

1. **Entity Import.** Two seed strategies: (a) Breadth-first search from Englishiki (Q1342448), seeding 500 entities then importing all their triples and linked entities. The BFS expansion produces **34,335 unique entities** (not 500), of which 1,781 contain diacritical marks. With aliases, the total embedding count reaches 41,725. (b) Broad P31 (instance of) sampling across country-level entities to provide a domain-general baseline. Both seeds contribute to the relational displacement analysis (Section 4.1); the collision analysis (Section 5.4) focuses on the Englishiki seed because its 1,781 diacritic-bearing labels trigger tokenizer collisions at scale.

2. **Embedding.** Each entity's English label is embedded using mxbai-embed-large (1024-dim) via Ollama. Aliases receive separate embeddings. Total: 41,725 embeddings from the Englishiki seed. Labels are short text strings (typically 1-5 words), consistent with how these models are used in practice for entity linking and retrieval.
3. **Relational Displacement Computation.** For each entity-entity triple, compute the displacement vector between subject and object label embeddings. Total: 16,893 entity-entity triples across 1,472 unique predicates. This is the standard $h + r \approx t$ test from TransE, applied without training.

3.3 Discovery Procedure

For each predicate p with ≥ 10 entity-entity triples:

1. Compute all relational displacements $\{g_i\}$
2. Compute mean displacement d_p
3. Compute consistency: mean alignment of each g_i with d_p
4. Compute pairwise consistency: mean cosine similarity between all pairs of displacements
5. Compute magnitude coefficient of variation: stability of displacement magnitudes

Note on unit-norm embeddings. mxbai-embed-large returns L2-normalized embeddings ($\|v\| = 1.0000$). Consequently, displacement magnitudes are a deterministic function of cosine similarity: $\|f(o) - f(s)\| = \sqrt{2(1 - \cos(f(o), f(s)))}$. The MagCV metric therefore carries no information independent of cosine distance for this model. We retain it for cross-model comparability, as other models (e.g., BioBERT) do not necessarily normalize.

3.4 Prediction Evaluation

For each discovered operation (consistency > 0.5), we evaluate prediction accuracy using **leave-one-out**:

For each triple (s, p, o) : 1. Compute $d_p^{(-i)} = \text{mean displacement excluding this triple}$ 2. Predict: $\hat{o} = f(s) + d_p^{(-i)}$ 3. Rank all entities by cosine similarity to $\hat{o}$ 4. Record the rank of the true object o

We report Mean Reciprocal Rank (MRR) and Hits@k for $k \in \{1, 5, 10, 50\}$.

3.5 Composition Test

To test whether operations can be chained, we find all two-hop paths $s \xrightarrow{p_1} m \xrightarrow{p_2} o$ where both p_1 and p_2 are discovered operations. We predict:

$$\hat{o} = f(s) + d_{p_1} + d_{p_2}$$

and evaluate whether the true o appears in the top-k nearest neighbors. We test 5,000 compositions.

4 Results

4.1 Operation Discovery

Of 159 predicates with ≥ 10 triples, 86 (54.1%) produce consistent displacement vectors:

Category	Count	Alignment Range
Strong operations	32	> 0.7
Moderate operations	54	$0.5 - 0.7$
Weak/no operation	73	< 0.5

Table 1. Distribution of discovered operations by consistency.

The top 15 discovered operations:

Predicate	Label	N	Alignment	Pairwise	MagCV	Cos Dist
P8324	funder	25	0.930	0.859	0.079	0.447
P2633	geography	18	0.910	0.819	0.097	0.200
P9241	demographics	21	0.899	0.799	0.080	0.215
P2596	of topic	16	0.896	0.790	0.063	0.202
P5125	culture	20	0.887	0.777	0.089	0.196
P7867	Wikimedia	29	0.878	0.763	0.099	0.205
P8744	outline	30	0.870	0.749	0.094	0.182
P1740	category	18	0.862	0.728	0.121	0.266
P1791	for maps	13	0.857	0.714	0.121	0.302
P1465	economy	29	0.857	0.725	0.124	0.249
P163	of topic	31	0.855	0.723	0.123	0.208
P2746	cat. for	11	0.850	0.696	0.048	0.411
P1923	films	32	0.831	0.681	0.042	0.387
P1464	shot here	32	0.814	0.653	0.145	0.265
P237	cat. for	21	0.798	0.620	0.138	0.268
	people					
	buried					
	here					
	cat. for					
	people					
	who died					
	here					
	flag					
	production					
	statistics					
	participating					
	team					
	cat. for					
	people					
	born here					
	coat of					
	arms					

Table 2. Top 15 relations by displacement consistency (alignment with mean displacement). N = number of triples. Pairwise = mean cosine similarity between all pairs of displacements. MagCV = coefficient of variation of displacement magnitudes. Cos Dist = mean cosine distance between subject and object.

4.2 Prediction Accuracy

Leave-one-out evaluation of all 86 discovered operations:

Predicate	Label	N	Align	MRR	H@1	H@10	H@50
P9241	demographics	21	0.899	1.000	1.000	1.000	1.000
P2596	of topic	16	0.896	1.000	1.000	1.000	1.000
P7867	culture	29	0.878	1.000	1.000	1.000	1.000
P8744	category	30	0.870	1.000	1.000	1.000	1.000
P5125	for maps	20	0.887	0.975	0.950	1.000	1.000
	economy						
	of topic						
	Wikimedia						
	outline						

Predicate	Label	N	Align	MRR	H@1	H@10	H@50
P2633	geography of topic	18	0.910	0.972	0.944	1.000	1.000
P1465	cat. for people who died here	29	0.857	0.966	0.966	0.966	0.966
P163	flag	31	0.855	0.937	0.903	0.968	1.000
P8324	funder	25	0.930	0.929	0.920	0.960	0.960
P1464	cat. for people born here	32	0.814	0.922	0.906	0.938	0.938
P237	coat of arms	21	0.798	0.858	0.762	0.952	1.000
P21	sex or gender	91	0.674	0.422	0.121	0.945	0.989
P27	country of citizenship	37	0.690	0.401	0.162	0.892	0.973

Table 3. Prediction results for selected operations (full table in supplementary). MRR = Mean Reciprocal Rank. H@k = Hits at rank k. The four predicates achieving MRR = 1.000 are functional predicates with highly consistent Wikidata naming conventions (e.g., every country has exactly one "Demographics of [Country]" article). Perfect MRR is expected when: (a) the predicate is strictly functional (one object per subject), (b) the displacement is consistent (alignment > 0.87), and (c) the object label is semantically close to a predictable transformation of the subject. Crucially, the string overlap null model (Section 4.4) confirms this is not a string manipulation artifact: these same predicates achieve string MRR of only 0.008–0.046 vs. vector MRR of 1.000. The embedding captures the semantic operation; the label convention merely makes the target unambiguous among 41,725 candidates.

Aggregate statistics across all 86 operations:

Metric	Value	95% Bootstrap CI
Mean MRR	0.350	—
Mean Hits@1	0.252	—
Mean Hits@10	0.550	—
Mean Hits@50	0.699	—
Correlation (alignment ↔ MRR)	r = 0.861	[0.773, 0.926]
Correlation (alignment ↔ H@1)	r = 0.848	[0.721, 0.932]
Correlation (alignment ↔ H@10)	r = 0.625	[0.469, 0.760]
Effect size: strong vs moderate MRR (Cohen's d)	3.092	(large)

Table 4. Aggregate prediction statistics with bootstrap confidence intervals (10,000 resamples). All correlations survive Bonferroni correction across 3 tests (adjusted alpha = 0.017).

The correlation between displacement consistency and prediction accuracy ($r = 0.861$, 95% CI [0.773, 0.926]) is practically useful as a quality filter. We note that this correlation has a natural mathematical component: when displacement variance is low (high consistency), the mean displacement is by construction a better predictor. However, the correlation is not fully tautological: consistency is computed over all triples, while MRR uses **leave-one-out** evaluation where each prediction excludes the test triple, and a high-consistency predicate could still have poor MRR if the predicted region is crowded with non-target entities. The effect size between strong (>0.7) and moderate (0.5–0.7) operations is Cohen's $d = 3.092$, indicating the 0.7 threshold cleanly separates high-performing from marginal operations.

4.3 Two-Hop Composition

Over 5,000 tested two-hop compositions ($S + d_1 + d_2$):

Metric	Value
Hits@1	0.058 (288/5000)
Hits@10	0.283 (1414/5000)
Hits@50	0.479 (2396/5000)
Mean Rank	1029.8

Table 5. Two-hop composition results.

Selected successful compositions ($\text{Rank} \leq 5$):

Chain	Rank
Tadahira $\rightarrow$ [citizenship] $\rightarrow$ Japan $\rightarrow$ [history of topic] $\rightarrow$ history of Japan	1
Tadahira $\rightarrow$ [citizenship] $\rightarrow$ Japan $\rightarrow$ [flag] $\rightarrow$ flag of Japan	1
Tadahira $\rightarrow$ [citizenship] $\rightarrow$ Japan $\rightarrow$ [cat. people buried here] $\rightarrow$ Category:Burials in Japan	2
Tadahira $\rightarrow$ [citizenship] $\rightarrow$ Japan $\rightarrow$ [cat. people who died here] $\rightarrow$ Category:Deaths in Japan	2
Tadahira $\rightarrow$ [citizenship] $\rightarrow$ Japan $\rightarrow$ [cat. associated people] $\rightarrow$ Category:Japanese people	3
Tadahira $\rightarrow$ [citizenship] $\rightarrow$ Japan $\rightarrow$ [head of state] $\rightarrow$ Emperor of Japan	4
Tadahira $\rightarrow$ [sex or gender] $\rightarrow$ male $\rightarrow$ [main category] $\rightarrow$ Category:Male	5

Table 6. Successful two-hop compositions. Note: all examples involve Fujiwara no Tadahira because our dataset is seeded from Englishiki (Q1342448), a Japanese historical text. Tadahira is one of the most densely connected entities in this neighborhood, appearing in many two-hop paths. The composition mechanism itself is general — the examples reflect dataset composition, not a limitation of the method.

4.4 String Overlap Null Model

A potential concern is that the discovered displacements merely capture string-level patterns — e.g., the displacement for "history of topic" (P2184) might simply encode the string prefix "History of" rather than relational knowledge. We test this with a string overlap null model: for each triple (s, p, o) , we rank all entities by longest common substring ratio with the subject label. If string overlap achieves comparable MRR to vector arithmetic, the displacement is trivially explained by surface patterns.

Result: Vector arithmetic outperforms string overlap in 39/39 tested predicates (100%). No predicate is trivially string-based.

Metric	Vector Arithmetic	String Overlap (LCS)	Token Overlap
Mean MRR	0.633	0.013	0.056
Predicates with MRR > 0.5	24	0	0

The gap is not marginal: mean vector MRR is $49\times$ higher than string MRR. Even the strongest string overlap scores (max 0.093 for P163 "flag") are far below the corresponding vector MRR (0.937).

The 24 predicates with vector MRR > 0.5 all have string MRR < 0.1, confirming that the embedding captures relational structure that cannot be recovered from label text alone.

Limitations of this baseline. The string overlap null model is deliberately simple — it tests whether vector arithmetic reduces to substring matching, not whether it outperforms all possible string-based methods. A more sophisticated baseline (e.g., regex pattern matching for predicates like "Demographics of [X]", or edit-distance heuristics) would likely close some of the gap for the most formulaic predicates. The $49\times$ ratio should be interpreted as evidence that the displacement is not a trivial string artifact, not as a claim about the difficulty of the prediction task itself. For the most formulaic predicates (demographics-of, geography-of), the prediction is easy by any method — the interesting finding is that vector arithmetic also works for predicates without formulaic naming (flag, coat of arms, head of state).

4.5 Failure Analysis

Predicates that resist vector encoding:

Predicate	Label	N	Alignment	Pattern
P3373	sibling	661	0.026	Symmetric
P155	follows	89	0.050	Sequence (variable direction)
P156	followed by	86	0.053	Sequence (variable direction)
P1889	different from	222	0.109	Symmetric/diverse
P279	subclass of	168	0.118	Hierarchical (variable depth)
P26	spouse	138	0.135	Symmetric
P40	child	254	0.142	Variable direction
P47	shares border with	197	0.162	Symmetric
P530	diplomatic relation	930	0.165	Symmetric
P31	instance of	835	0.244	Too semantically diverse

Table 7. Predicates with lowest consistency. Pattern = our characterization of why the displacement is inconsistent.

Three failure modes emerge:

1. **Symmetric predicates** (sibling, spouse, shares-border-with, diplomatic-relation): No consistent displacement direction because $f(A) - f(B)$ and $f(B) - f(A)$ are equally valid. Alignment ≈ 0 .
2. **Sequence predicates** (follows, followed-by): The displacement from "Monday" to "Tuesday" has nothing in common with the displacement from "Chapter 1" to "Chapter 2." The *relationship type* is consistent but the *direction in embedding space* is domain-dependent.
3. **Semantically overloaded predicates** (instance-of, subclass-of, part-of): "Tokyo is an instance of city" and "7 is an instance of prime number" produce wildly different displacement vectors because the predicate covers too many semantic domains.

Instance-of (P31) at 0.244 is particularly notable. It is the most important predicate in Wikidata (835 triples in our dataset) and a cornerstone of first-order logic, yet it does not function as a vector operation. This suggests that embedding spaces systematically under-represent relational structure: the space encodes *entities* well but *predicates* poorly.

4.6 Cross-Model Generalization

To test whether discovered operations are model-agnostic or artifacts of a single model's training, we ran the full pipeline on two additional embedding models: nomic-embed-text (768-dim) and all-minilm (384-dim). All three models were given identical input: the same Wikidata entities seeded from Englishiki (Q1342448) with --limit 500.

Model	Dimensions	Embeddings	Discovered	Strong (>0.7)
mxbai-embed-large	1024	41,725	86	32
nomic-embed-text	768	69,111	101	54
all-minilm	384	54,375	109	41

Table 8. Operations discovered per model. All three models discover operations despite different architectures and dimensionalities.

30 operations are universal — discovered by all three models. These include demographics-of-topic (avg alignment 0.925), culture (0.923), economy-of-topic (0.896), flag (0.883), coat of arms (0.777), and central bank (0.793). The universal operations are exclusively functional predicates, confirming the functional-vs-relational split across architectures.

Overlap Category	Count
Found by all 3 models	30
Found by 2 models	15
Found by 1 model only	30

Table 9. Cross-model operation overlap. 30 universal operations constitute the model-agnostic core.

Cross-model consistency correlations (alignment scores on shared predicates): mxbai vs all-minilm $r = 0.779$, mxbai vs nomic $r = 0.554$, nomic vs all-minilm $r = 0.358$. The positive correlations confirm that consistency is not random — predicates that work well in one model tend to work well in others, though the strength varies by model pair.

The same relational structure emerges across three unrelated embedding models with different architectures, different dimensionalities, and different training data. The discovered operations are properties of the semantic relationships themselves, not artifacts of any particular model.

5 Discussion

5.1 Relation Types and Displacement

The pattern across Tables 2 and 7 confirms what the KGE literature predicts: **consistent displacements emerge for functional (many-to-one) and bijective (one-to-one) relations, and fail for symmetric, transitive, or many-to-many relations.** Each country has one flag, one coat of arms, one head of state — these produce consistent displacements. Symmetric relations (sibling, spouse, shares-border-with) produce no consistent direction because $f(A) - f(B)$ and $f(B) - f(A)$ are equally valid.

That this pattern holds in general-purpose text embedding models — models with no relational training signal — confirms that the relational structure is a property of the semantic relationships themselves. Any embedding model that captures semantic similarity will encode functional relations as consistent displacements and fail on symmetric ones.

5.2 The Consistency-Accuracy Correlation

The $r = 0.861$ correlation between consistency and prediction accuracy is useful as a practical quality indicator but should not be overstated. There is a natural mathematical tendency for low-variance displacement vectors (high consistency) to produce better mean-based predictions — if all displacements point roughly the same direction, the mean will be a good predictor almost by construction. The correlation is therefore partly a geometric property of high-dimensional spaces, not purely an empirical discovery about these specific embedding models. What *is* empirically informative is the magnitude of the effect size between strong and moderate operations (Cohen's $d = 3.092$), which suggests the consistency threshold at 0.7 cleanly separates operations that work well from those that do not. The correlation is practically useful as a quality filter, even if its theoretical status is less remarkable than "self-diagnostic" framing might suggest.

5.3 Collision Geography

We independently measure two properties of each embedding: (a) its local density (mean k-NN distance) and (b) whether it collides with a semantically distinct entity at cosine ≥ 0.95 . Dense regions could in principle have few collisions if the model separates semantically distinct entities effectively even in crowded neighborhoods. The following results describe what we observe when diacritic-rich input is embedded.

5.4 The Embedding Collapse: [UNK] Token Dominance in mxbai-embed-large

A previously unreported defect in a widely-used model. mxbai-embed-large is one of the most popular open-source embedding models, with widespread deployment in RAG systems, semantic search, and knowledge graph applications. Despite this adoption, the tokenizer defect we report here — affecting over 16,000 entities and producing 147,687 colliding embedding pairs — appears to have gone undetected. Standard embedding benchmarks (MTEB, etc.) do not systematically probe non-Latin or diacritic-rich inputs at scale; a BFS traversal from a domain-specific seed does, because the knowledge graph naturally reaches the obscure terminology that benchmarks miss.

The Jinmyōchō collapse. Our collision analysis finds 147,687 cross-entity embedding pairs with cosine similarity ≥ 0.95 that represent genuine semantic collisions: different text mapped to near-identical vectors. This count reflects *pairwise* collisions: if k entities cluster together, they contribute $\binom{k}{2}$ pairs. The 147,687 total arises from approximately 16,067 entities (of 41,725) participating in at least one collision, organized into clusters of varying size. "Jinmyōchō" collides with 504 unique texts spanning romanized Japanese (kugyō, Shōtai), Arabic (Djazaïr, Filastīn), Irish (Éire), Brazilian indigenous languages (Aikanã, Amanayé), and IPA characters — words that share no orthographic or semantic relationship whatsoever.

The mechanism is [UNK] token dominance, not diacritic stripping. An earlier version of this paper attributed the collisions to "WordPiece diacritic stripping," but this explanation is incomplete and misleading. If the tokenizer simply stripped diacritics, "Hokkaidō" would become "Hokkaido" and "Djazaïr" would become "Djazair" — different strings that should produce different embeddings. The actual mechanism is more severe:

1. mxbai-embed-large uses a WordPiece tokenizer whose vocabulary does not include characters with diacritical marks (ō, ū, ī, ï, ș, ț, é, â, etc.).
2. When the tokenizer encounters these out-of-vocabulary characters, it replaces them with the [UNK] (unknown) token.
3. For short input strings where diacritical characters constitute a significant fraction of the content, the tokenized sequence becomes dominated by [UNK] tokens.
4. The BERT-based embedding model pools over this [UNK]-dominated sequence, producing an embedding that reflects the [UNK] token's learned representation rather than the actual text content.
5. **All short diacritical strings converge to the same [UNK]-dominated attractor region,** regardless of language, script, or meaning.

Controlled evidence. We embed test pairs to confirm the mechanism (full data in collisions.csv):

Pair	Cosine Similarity	Interpretation
"Hokkaidō" ↔ "Éire"	1.000	Different languages, different meanings — identical embedding
"Jinmyōchō" ↔ "Filastīn"	1.000	Japanese ↔ Arabic — identical embedding
"Djazaïr" ↔ "România"	1.000	Arabic ↔ Romanian — identical embedding
"naïve" ↔ "Zürich"	1.000	French ↔ German — identical embedding

Pair	Cosine Similarity	Interpretation
"Hokkaidō" ↔ "Hokkaido"	0.450	Same word, diacritic vs. ASCII — dissimilar
"Tōkyō" ↔ "Tokyo"	0.500	Same word, diacritic vs. ASCII — dissimilar
"Tokyo" ↔ "Berlin"	0.751	Control: two capitals — normal similarity

Table 10. Controlled collision pairs. The diacritical version of a word is more similar to an unrelated diacritical word in a different language (cosine 1.0) than to its own ASCII equivalent (cosine ~0.45). This rules out diacritic stripping as the mechanism: if the model stripped diacritics and embedded the ASCII form, "Hokkaidō" would be close to "Hokkaido", not to "Éire". Instead, the [UNK] tokens overwhelm the embedding, and all [UNK]-dominated inputs converge to the same point.

The collapse zone is dense, not sparse. Geometric analysis of 16,067 colliding embeddings (vs. 74,760 non-colliding) reveals:

1. **Colliding embeddings are $2.4\times$ denser than non-colliding ones.** Mean k-NN distance for colliding embeddings is 0.106, vs 0.258 for non-colliding (ratio $0.41\times$).
2. **71% of colliding embeddings fall in the densest quartile**, vs the expected 25% if uniformly distributed. Only 3.2% fall in the sparsest quartile.
3. **The collapse zone is not geometrically isolated.** The distance from a colliding embedding to its nearest non-colliding neighbor (mean 0.119) is nearly identical to the non-colliding-to-non-colliding distance (mean 0.121, ratio $0.98\times$).

This means the [UNK] attractor region sits *among* the well-structured embeddings, not apart from them. The colliding embeddings crowd into already-dense neighborhoods where the model cannot differentiate them from legitimate nearby entities.

The defect is silent and likely exploitable. The [UNK]-dominated embedding region has several concerning properties: (1) it is invisible to standard benchmarks, (2) the model returns a confident-looking embedding vector rather than an error, (3) any downstream system treating this vector as meaningful will silently produce wrong results. We hypothesize that this represents a class of silent failures that has been causing degraded performance in production systems for years — any RAG pipeline, semantic search engine, or knowledge graph application processing non-ASCII input through mx-bai-embed-large has been mapping those inputs to a single undifferentiated region. The scale of affected systems is difficult to estimate, but given the model's popularity and the prevalence of diacritical marks in non-English text, the impact is likely substantial.

The phenomenon is reminiscent of glitch tokens (Li et al., 2024) but at a different scale: entire *classes of input* (any text containing diacritical marks) rather than individual tokens, and in sentence-embedding models rather than LLMs.

Why the Englishiki seed matters. Englishiki (Q1342448) is a 10th-century Japanese text whose entities include romanized shrine names (Jinmyōchō, Shikinaisha), historical Japanese personal names, and linked entities from Arabic, Irish, and indigenous-language Wikipedia articles. This floods the embedding space with exactly the inputs that trigger [UNK] token dominance, making the phenomenon measurable at scale. The defect exists regardless of seed choice — any diacritical input triggers it — but the Englishiki seed makes it *statistically visible* by providing thousands of affected entities in a single BFS traversal.

5.5 Practical Implications

The [UNK] token dominance defect has immediate practical consequences. Any system using mx-bai-embed-large for semantic search, RAG, or knowledge graph completion over non-ASCII text has been silently affected. A user querying "Hokkaidō" will retrieve results from the [UNK] attractor region — potentially returning "Éire", "Djazaïr", or any other diacritical string — rather

than results related to the Japanese island. The failure is silent: the model returns a valid-looking 1024-dimensional vector, and no error is raised.

We hypothesize that this class of silent failure extends beyond mxbai-embed-large. All three models tested use WordPiece or similar subword tokenizers, and the [UNK] token dominance mechanism applies to any tokenizer with incomplete character coverage. The practical recommendation is to test embedding models with diacritic-rich input before deployment, and to prefer models with byte-level tokenizers (e.g., CANINE, ByT5) or SentencePiece with full Unicode coverage for multilingual applications.

5.6 Limitations

1. **Three embedding models.** We validate across mxbai-embed-large (1024-dim), nomic-embed-text (768-dim), and all-minilm (384-dim), finding 30 universal relations. All three are English-language text embedding models trained on similar corpora. Testing on multilingual models or domain-specific models (e.g., biomedical) would further characterize the generality of the three-regime structure.
2. **Collision geometry analysis covers one seed.** The distance metrics characterizing the embedding collision zone (Section 5.4) are computed from the Englishiki-seeded dataset. Multi-seed analysis would test whether the same crowding pattern holds across domains.
3. **Label embeddings only.** We embed entity *labels* (short text strings), not descriptions or full articles. This deliberately mirrors how these models are used in practice for entity linking and knowledge graph completion (short query strings, not full documents). Richer textual representations might shift some entities out of the sparse zone, but the label-only setting represents a common real-world deployment pattern for these models.
4. **Potential training data overlap.** The embedding models tested were trained on large web crawls that likely include Wikipedia content, and Wikidata entities often have corresponding Wikipedia articles. This raises the possibility that some discovered displacements reflect memorized associations from training data rather than emergent geometric structure. The cross-model consistency (30 universal operations across three independently trained models) provides partial mitigation: memorization patterns would be model-specific, while consistent operations across architectures suggest structural encoding. However, a definitive test would require embedding models trained on corpora that exclude Wikipedia, which we leave for future work.
5. **Single tokenizer family.** All three models use WordPiece or similar subword tokenizers. The [UNK] token dominance mechanism is specific to tokenizers with incomplete character coverage. Models using byte-level tokenizers (e.g., CANINE, ByT5) or SentencePiece with full Unicode support would likely not exhibit this specific failure mode. Testing against such models would clarify the scope of the defect.
6. **Relational displacement, not full FOL.** We test which binary relations encode as consistent vector arithmetic. Full first-order logic includes quantifiers, variable binding, negation, and complex formula composition, none of which we test. Extending the displacement analysis to richer logical operations is future work.

6 Conclusion

We apply latent space cartography — systematic relational displacement analysis using knowledge graph triples — to three general-purpose text embedding models. The procedure, which packages standard TransE-style evaluation into a replicable pipeline, identifies 30 relations that manifest as consistent vector displacements across all three models. The functional-vs-symmetric split predicted by the KGE literature reproduces across models and domains.

The primary finding is a silent defect in mxbai-embed-large caused by [UNK] token dominance in the WordPiece tokenizer. Characters with diacritical marks are absent from the tokenizer vocabulary, replaced with [UNK] tokens, and when these tokens dominate short input strings, the embedding converges to a single attractor region regardless of text content. Controlled pairs confirm the mechanism: the diacritical version of a word is more similar to an unrelated diacritical word in a different language (cosine 1.0) than to its own ASCII equivalent (cosine ~ 0.45). This defect affects 16,067

entities in our dataset (147,687 colliding pairs), is concentrated in the densest regions of the embedding space, and is invisible to standard benchmarks. It has existed for years in a widely-deployed model, silently degrading any downstream system processing non-ASCII input.

The defect was discovered because the cartographic procedure, seeded from a Japanese historical text (Engishiki), naturally reached the diacritic-rich terminology that standard benchmarks never test. This suggests a broader lesson: systematic probing of embedding spaces with domain-specific knowledge graphs can surface defects that generic benchmarks miss. The practical recommendation is to test embedding models with representative non-ASCII input before deployment.

All code and data are publicly available.

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